Musical Metrics

musicalmetrics.xyz

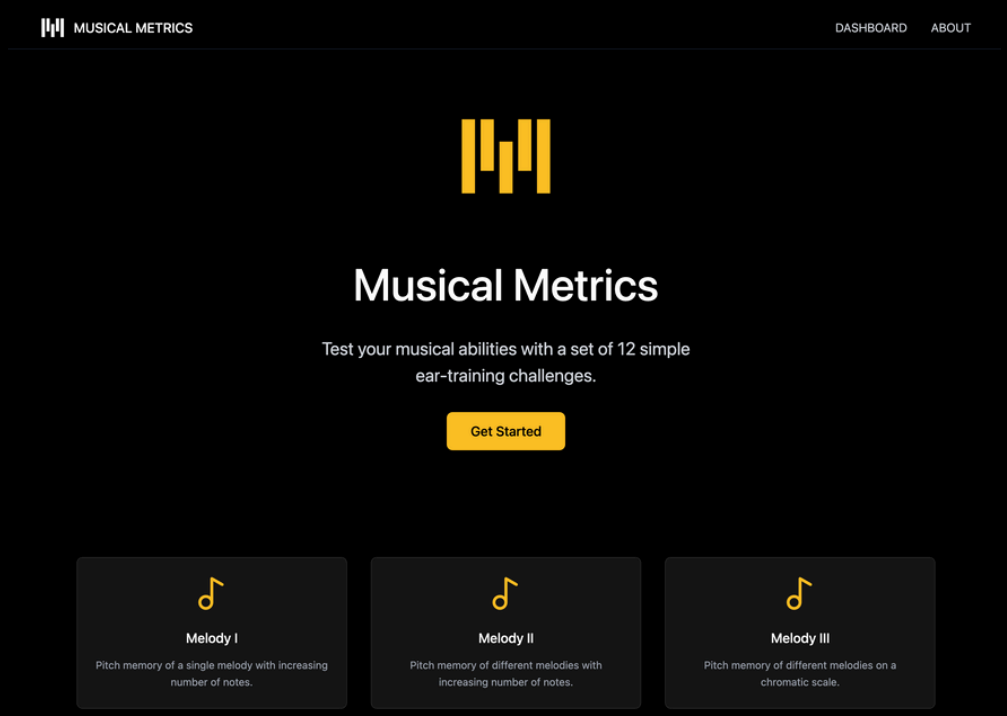
Start Playing

Musical Metrics is a set of 12 simple and creative web-based music games that tests a player's listening ability in various aspects of music and hearing, examining a diverse range of listening skills such as relative pitch and rhythmic recognition. The games are part of an ongoing research to explore hearing ability as a major component of a musician's learning progression.

The games are of various difficulty levels. Some are beginner-friendly and are accessible to people with no musical experience, while some test the capability of even the most proficient musicians. A few games require a certain level of understanding of musical theory and terminology, but are nonetheless playable after a few practices.

Musical Metrics isn't only a useful tool for testing players' hearing ability, but also a great method of ear-training, all while having fun, whether that's competing with friends and family, or going against other musicians.

Concept & Implementation



The concept of Musical Metrics was born when I became interested in how listening ability contributes to musical learning. I had the idea of designing simple games that would allow musicians to systematically evaluate different musical aspects of their hearing, and exploring whether such games would have educational benefits. The first game, “Melody I”, was created to test relative pitch. My research idea was to investigate whether a person’s number of years playing an instrument has a direct correlation with performance on these games.

Now at 12 games, Musical Metrics is made entirely with HTML and JavaScript, while the UI was modelled in Figma. Currently, the game only runs on frontend; the next step is to implement the backend, in order to enable features such as saving high scores, collecting data from global users, and displaying a leaderboard.

Game Overview

Melody I

Pitch memory of a single melody with increasing number of notes.

Melody II

Pitch memory of different melodies with increasing number of notes.

Melody III

Pitch memory of different melodies on a chromatic scale.

Interval I

Musical interval recognition between two rising notes in the same octave.

Interval II

Musical interval recognition with extended range and compound intervals.

Harmony I

Distinguish between six basic chord types.

Harmony II

Increased range of chord types and extensions.

Harmony III

Complex harmony recognition with inversions.

Tempo I

Do you have perfect tempo? Find out with this simple BPM recognition game.

Tempo II

How well can you maintain your internal tempo? Find out by tapping back the metronome beats.

Pitch I

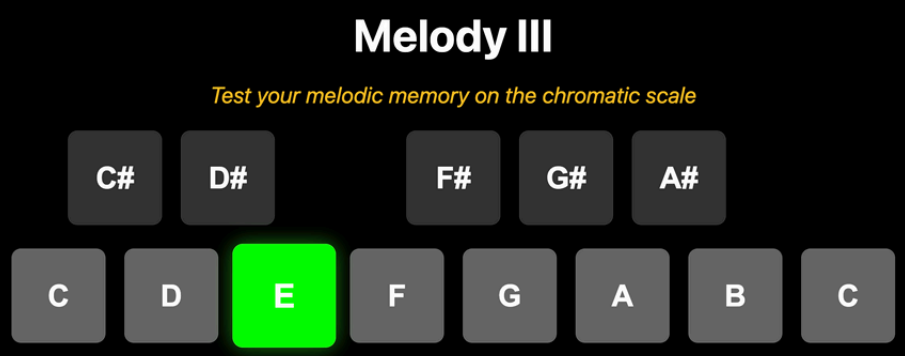
Test your pitch detection skills by identifying cents deviations from reference notes.

Rhythm I

Master basic rhythmic patterns by tapping random beats containing quavers, triplets, semiquavers and sextuplets.

The games in Musical Metrics are currently divided into six sub-categories: Melody, Interval, Harmony, Tempo, Pitch, and Rhythm. For those categories with multiple games, the difficulty increases with each level, as will be illustrated in the pages below. Some games, such as Melody, end when the player makes a mistake; while the others, realising that a proficient player would likely never make an error, are limited to a specific number of rounds. Nevertheless, each game is challenging in its own way.

MELODY



Melody is the first creation of Musical Metrics. It tests the relative pitch and melodic memory of a player. For all three games, a melody of an increasing length is played in each level by the computer, and the player is asked to play back the exact melody upon hearing it. Some level of musical experience is required for these three games.

- Melody I is the most beginner-friendly version. Level 1 starts on a single random note (in the C major scale), and each subsequent level adds one note to the note before, gradually building a melody. Players benefit from hearing the melody repeatedly.
- Melody II is similar to Melody II except for that in this version, every level introduces a new, random melody. This increases the difficulty substantially.
- Melody III is among the most difficult of all Musical Metrics games. It functions like Melody II, yet on a chromatic scale, which makes it almost impossible for players only used to hearing diatonic melodies.

INTERVAL

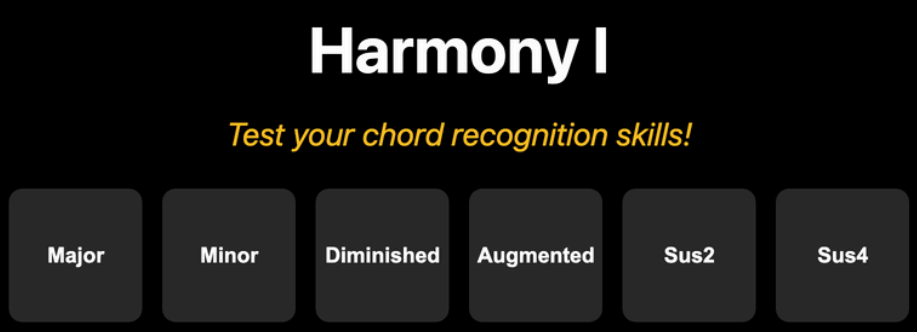
Interval I

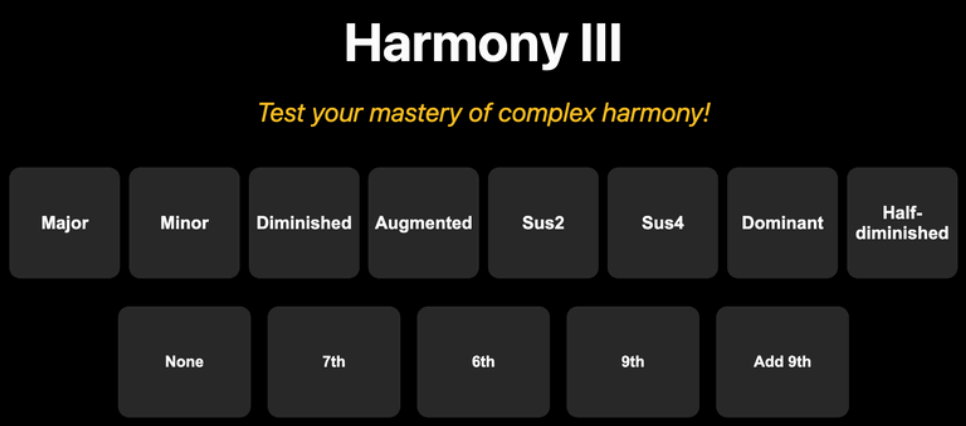
Test your interval recognition skills!

Interval requires certain knowledge of music theory. In Interval, two notes are played at random one after another, and the player is asked to identify the musical interval heard. In each round, they must correctly select both the quality (major, minor, etc.) and the distance (2nd-8th) to score a point.

- Interval I is an accessible version. The notes played are always selected from the same octave range (C4-C5), and the first note played is always middle C. In this case, the player only needs to identify the second note in order to find the correct answer.
- Interval II introduces several twists: the game can play any notes in a three-octave range, meaning that compound intervals are inevitably a feature (however, the player is asked to select the simple interval equivalent when they hear compound intervals). Furthermore, there is no standard “first note”, and the higher pitched note can be played first, making interval identification much more difficult.

HARMONY

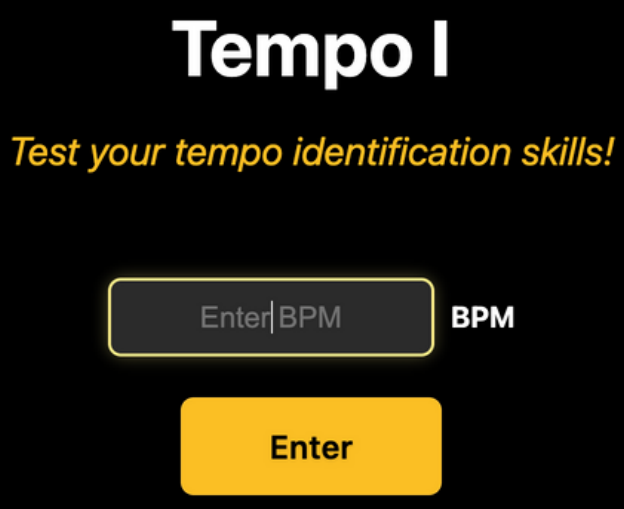


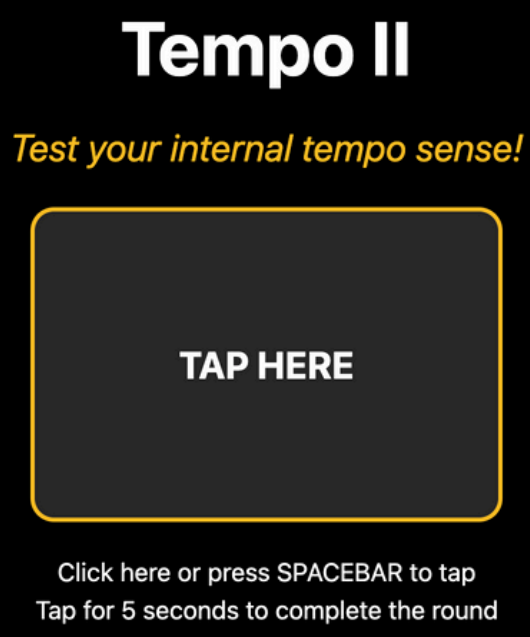


Harmony is a set of games with a very simple concept: the computers plays a chord at random, and the player is asked to select the correct type of chord heard.

- Harmony I only asks the player to distinguish between six basic chord types. Chords are always played in their root positions, making it easier for the players. One observation, though, is that when sus2 and sus4 chords are isolated on their own, they might be tricky to distinguish.
- Harmony II introduces two new types of chords above that of Harmony I: dominant and half-diminished. Furthermore, 7th chords are added to introduce a new layer of difficulty. The player now not only has to select the main chord type, but also whether it is a 7th.
- Harmony III not only introduces inversions, but also adds the possibilities of 6th, 9th, and add 9th chords.

TEMPO





People with “perfect tempo” are much rarer than those with perfect pitch. For those who do have perfect tempo, **Tempo I** would be pure cheating. Yet, for the rest, it would be a great way to train their internal metronome and sense of tempo.

- Tempo I is designed simply for the player to attempt to identify the BPM at which a random metronome beat is playing. It could be difficult for some but rather easy for the others. There is, however, a workaround that improves accuracy: if you start counting seconds, that will be 60 BPM, and you just work your way from there.
- Tempo II is very easy to play, but demands a different skill: the ability to keep the same tempo going. In each round, the computer plays a random BPM, and the player is asked to tap that tempo. Most people will find this an easy exercise, and score decently well.

PITCH

Pitch I

Test your pitch detection skills!

Reference Note: G#3

Play Reference

Play Test Note

Enter cents

cents

Submit

Pitch I is the most unique game in Musical Metrics, in that it asks the player's ears to carry out unconventional listening (instrumentalists who tune their instrument will probably find it easier, though). The game plays a reference note, and then the player is asked to listen to the test note, which makes the reference note out of tune by a maximum of 50 cents sharp or flat. To implement the pitch deviation, I used Tone.js, a popular Web Audio framework. The player is then required to enter the amount of pitch deviation they believe was present. Although an average human can distinguish between 5-10 cents differences in pitch, stating the exact amount is extremely hard.

RHYTHM

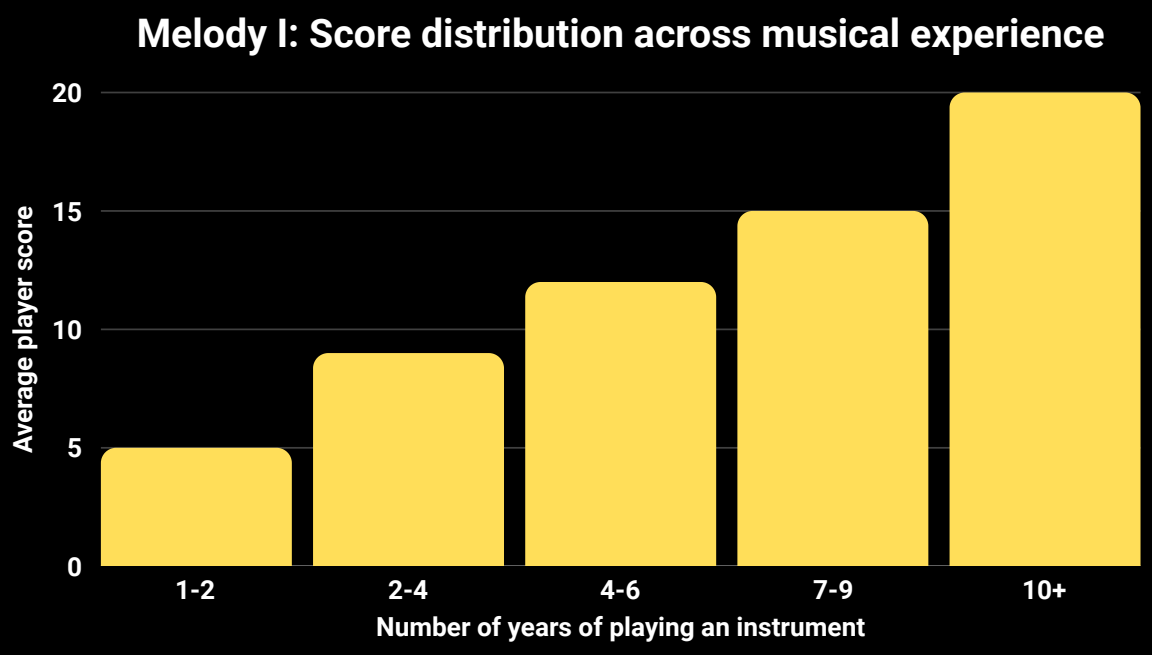
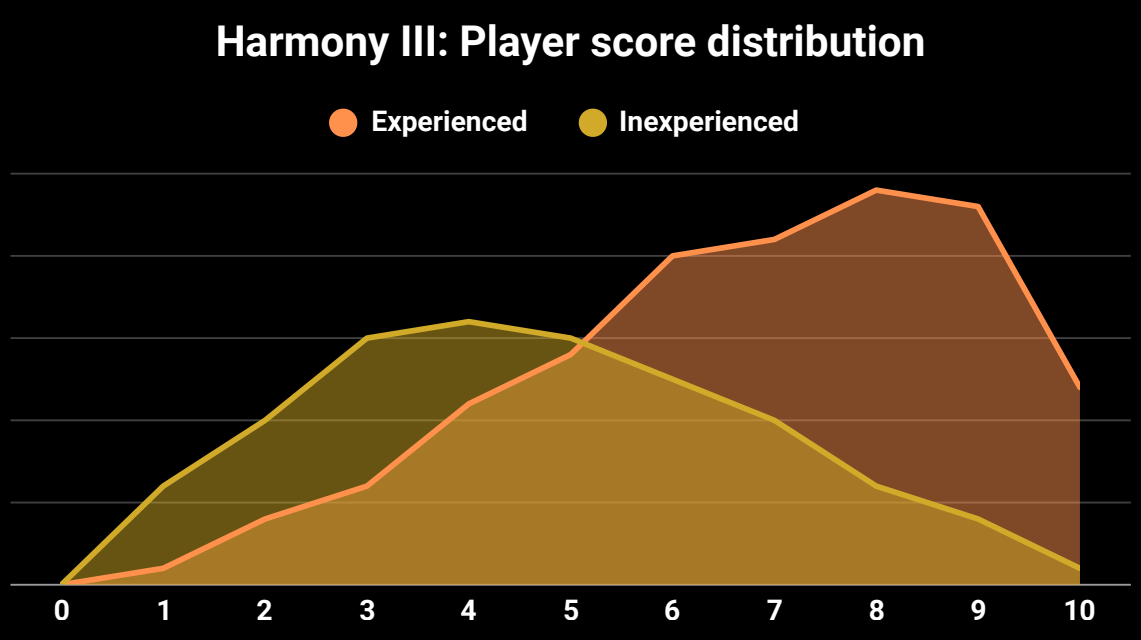
Rhythm I

Test your rhythmic accuracy!

Click here or press SPACEBAR to tap
Tap the rhythm pattern you heard

Taps: 0 / 14

Rhythm I, being the newest addition to Musical Metrics, presents its own unique challenges. In each round, the computer plays a 4-beat rhythm. During each beat, it randomly decides between playing quaver, triplet, semiquaver or sextuplets. The player then taps back the exact rhythm as accurately as possible, although tapping at a different tempo is not penalised, as long as the relative rhythm is correct. It is recommended that players try to identify the notes that comprise the rhythm (quaver, triplet, etc.), before tapping it back. This would yield much superior results than simply attempting to match the speed of the computer.

Research & Goals

Below are some examples of data collection & processing made possible by Musical Metrics, which can be useful statistics for my research. This can be further enhanced in the future by incorporating global user data.